

Modeling Loci and Intersections with Conic Sections

Answer Key

Algebra 2

Quick Answers

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|--|--|
| 1. $x^2 + y^2 - 6x + 4y - 12$ | (b) radius = 4 |
| 2. 3 | 8. $x = 3, y = 4, x = -4, y = -3$ |
| 3. -2, 2 | 9. (a) y when $x = 2 = -4, 4, —$ |
| 4. 5 | 10. (a) simplified equation = $x^2 + y^2 - 16, —$ |
| 5. (a) the value of $4p = 20$, (b) x when $y = 5 = -10, 10$ | 11. $x = 4, y = 3, x = -4, y = 3, x = 3, y = -4, x = -3, y = -4$ |
| 6. $x = 3, y = 6, x = -1, y = -2$ | 12. (a) values of $k = 4, -4, —$ |
| 7. (a) simplified equation = $x^2 + y^2 - 16x + 48$, | |
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What is verified

14 of 17 answers machine-checked against SymPy, across 12 problems.

— marks an answer only you can judge — problems 9, 10, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSA-REI.C.7 – problems 3, 6, 8, 11, 12

HSG-GPE.A.1 – problems 1, 4, 7, 10

HSG-GPE.A.2 – problems 2, 5, 9

Difficulty 1–5 of 5 across 17 tagged checks.

1. P moves so that $PC = 5$ with $C(3, -2)$. Write the expanded equation.

Solution: “Distance from C is 5” is the definition of a circle, so square the distance formula:

$$\begin{aligned}(x - 3)^2 + (y + 2)^2 &= 25 \\ x^2 - 6x + 9 + y^2 + 4y + 4 &= 25\end{aligned}$$

Collecting the constants gives

$x^2 + y^2 - 6x + 4y - 12 = 0$

2. $x^2 = 12y$ (focus $(0, 3)$, directrix $y = -3$). Find y when $x = 6$.

Solution: $36 = 12y$, so $y = 3$. Notice the point $(6, 3)$ sits level with the focus — its distance to the focus is 6 and its distance to the directrix is $3 - (-3) = 6$. $y = 3$

3. Where does $y = x^2$ meet $y = 4$?

Solution: Setting the two expressions for y equal gives $x^2 = 4$, and a horizontal line cuts the parabola twice.

$$\boxed{x = -2 \quad \text{and} \quad x = 2}$$

4. $x^2 + y^2 - 6x + 4y - 12 = 0$. Complete the square and state the radius.

Solution: Group and complete each square:

$$\begin{aligned}(x^2 - 6x + 9) + (y^2 + 4y + 4) &= 12 + 9 + 4 \\(x - 3)^2 + (y + 2)^2 &= 25\end{aligned}$$

The centre is $(3, -2)$ and the radius is $\sqrt{25}$.

$$\boxed{r = 5}$$

5. Vertex $(0, 0)$, focus $(0, 5)$, $x^2 = 4py$. **(a)** $4p$. **(b)** x when $y = 5$.

Solution (a): p is the directed distance from the vertex to the focus, so $p = 5$ and $4p = 20$. $4p = 20$

Solution (b): The equation is $x^2 = 20y$, so at $y = 5$: $x^2 = 100$. This chord through the focus has length 20, which is exactly $|4p|$.

$$\boxed{x = -10 \quad \text{and} \quad x = 10}$$

6. Intersect $y = x^2 - 3$ with $y = 2x$.

Solution: Substitute the line into the parabola:

$$\begin{aligned}x^2 - 3 &= 2x \\x^2 - 2x - 3 &= 0 \\(x - 3)(x + 1) &= 0\end{aligned}$$

Then $y = 2x$ gives the matching heights: $x = 3 \Rightarrow y = 6$, and $x = -1 \Rightarrow y = -2$.

$$\boxed{(3, 6) \quad \text{and} \quad (-1, -2)}$$

7. $PA = 2PB$ with $A(0, 0)$, $B(6, 0)$. **(a)** Simplify. **(b)** Radius.

Solution (a): Squaring the condition avoids the radicals: $PA^2 = 4PB^2$.

$$\begin{aligned}x^2 + y^2 &= 4[(x - 6)^2 + y^2] \\x^2 + y^2 &= 4x^2 - 48x + 144 + 4y^2 \\0 &= 3x^2 + 3y^2 - 48x + 144\end{aligned}$$

Dividing by 3 gives

$$x^2 + y^2 - 16x + 48 = 0$$

Solution (b): Complete the square: $(x - 8)^2 + y^2 = 64 - 48 = 16$. The centre is $(8, 0)$ — not at A , not at B , and not at their midpoint.

$$r = 4$$

8. Intersect $x^2 + y^2 = 25$ with $y = x + 1$.

Solution: Substitute the line into the circle:

$$\begin{aligned} x^2 + (x + 1)^2 &= 25 \\ 2x^2 + 2x - 24 &= 0 \\ x^2 + x - 12 &= 0 \\ (x - 3)(x + 4) &= 0 \end{aligned}$$

Back-substituting into $y = x + 1$: $x = 3 \Rightarrow y = 4$, and $x = -4 \Rightarrow y = -3$.

$$(3, 4) \quad \text{and} \quad (-4, -3)$$

9. $y^2 = 8x$ (focus $(2, 0)$, directrix $x = -2$). (a) y when $x = 2$. (b) Locus argument for $(2, 4)$.

Solution (a): $y^2 = 8(2) = 16$.

$$y = -4 \quad \text{and} \quad y = 4$$

Solution (b): *Instructor-judged.* The argument has to use the defining property. Distance from $(2, 4)$ to the focus $(2, 0)$ is 4; distance from $(2, 4)$ to the directrix $x = -2$ is $2 - (-2) = 4$. Equal distances is precisely what places a point on the parabola. Substituting into $y^2 = 8x$ checks the algebra rather than the definition and earns no credit for this part.

10. $PA^2 + PB^2 = 50$ with $A(-3, 0)$, $B(3, 0)$. (a) Simplify. (b) Describe the locus.

Solution (a): Write both squared distances and add:

$$\begin{aligned} (x + 3)^2 + y^2 + (x - 3)^2 + y^2 &= 50 \\ 2x^2 + 18 + 2y^2 &= 50 \\ 2x^2 + 2y^2 - 32 &= 0 \end{aligned}$$

Dividing by 2 gives

$$x^2 + y^2 - 16 = 0$$

Solution (b): *Instructor-judged.* A complete answer names all three features and points to the equation for each: it is a circle (the x^2 and y^2 coefficients match and are positive), centred at the origin (no x or y term survived), with radius 4 (since $x^2 + y^2 = 16$). The observation worth crediting is that A and B have disappeared from the answer: the cross terms cancelled, so the condition does not care which fixed point P is nearer. Shape alone, without centre and radius, is half credit.

11. Intersect $x^2 + y^2 = 25$ with $y = x^2 - 13$.

Solution: The parabola gives $x^2 = y + 13$; substitute that into the circle so the equation is in y alone:

$$\begin{aligned}(y + 13) + y^2 &= 25 \\ y^2 + y - 12 &= 0 \\ (y + 4)(y - 3) &= 0\end{aligned}$$

For $y = 3$: $x^2 = 16$, so $x = \pm 4$. For $y = -4$: $x^2 = 9$, so $x = \pm 3$. Each y supplies two points, which is why there are four.

$$\boxed{(4, 3), (-4, 3), (3, -4), (-3, -4)}$$

12. $y = x + k$ meets $x^2 + y^2 = 8$ at exactly one point. **(a)** Find k . **(b)** What one point means.

Solution (a): Substitute and collect:

$$\begin{aligned}x^2 + (x + k)^2 &= 8 \\ 2x^2 + 2kx + (k^2 - 8) &= 0\end{aligned}$$

One intersection means one repeated root, so the discriminant vanishes:

$$\begin{aligned}(2k)^2 - 4(2)(k^2 - 8) &= 0 \\ -4k^2 + 64 &= 0\end{aligned}$$

$$\boxed{k = -4 \quad \text{and} \quad k = 4}$$

Solution (b): *Instructor-judged.* Two links are needed. Algebraically, exactly one intersection means the quadratic in x has a repeated root, which is what a zero discriminant says. Geometrically, that is the line touching the circle at a single point — a tangent, and the two values of k are the tangent lines of slope 1 on either side of the centre. A response that computes the discriminant without saying what a repeated root means for the picture, or that says “tangent” without the discriminant, has made only one of the two links.